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Cross-Asset Volatility and Equity Risk

Empirical Report

Quantitative Techniques for Financial Economics
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Preface

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1. Research Question and Economic Motivation

1.1 Research Question and Economic Motivation

Mandelbrot (1963, p. 418) observed that "large changes tend to be followed by large changes, of either sign, or small changes tend to be followed by small changes." This persistence makes a market's own history a natural starting point for volatility forecasting. It also raises a question: can information from other markets improve forecasts beyond what is already captured by past equity volatility?

Forecasts of equity volatility inform risk measurement, portfolio allocation and derivative pricing, making forecast accuracy relevant to investors and financial institutions. For researchers, examining the contribution of other markets provides a way to assess whether cross-market relationships contain useful predictive information.

Commodity and Treasury markets offer plausible sources of such information. Uncertainty in energy markets may affect expectations about production costs, inflation and corporate profitability. Gold-market volatility may reflect changing perceptions of financial risk, while Treasury yield movements may capture shifts in expectations about interest rates and economic conditions. These connections motivate investigating whether developments in other markets help anticipate future equity variance (Yongdeng Xu, et. Al., 2024).

This study therefore asks whether incorporating information from commodity and US Treasury markets improves forecasts of S&P 500 futures realised variance relative to models relying solely on historical equity information. We examine commodity futures, including crude oil, gold and natural gas, alongside a Treasury-market proxy constructed from changes in the ten-year US yield.

The central issue is the incremental forecasting value of these additional inputs. Markets may respond simultaneously to common shocks without their histories providing useful signals about future equity variance. Adding variables also increases the number of parameters to estimate, so a broader information set does not automatically produce more accurate forecasts. We therefore compare equity-only models with specifications incorporating additional market information and evaluate their out-of-sample performance. Complementary modelling approaches, including HAR and VAR models, allow us to examine whether the findings depend on how volatility dynamics are represented. Our objective is to assess predictive usefulness rather than establish causal transmission between markets (Francis X, et. Al., 2012).

2. Data

We measure volatility by realised variance, which sums squared intraday returns and therefore measures daily variance more precisely than squared daily returns, providing a reliable target for forecast evaluation (Andersen et al., 2003). We use the five-minute version (RV5), which Liu et al. (2015) find difficult to outperform across asset classes. Sampling more frequently would add market microstructure noise, while sampling less frequently would discard information.

All markets are represented by liquid futures contracts, most of which trade almost continuously on CME Group exchanges, so their variances cover comparable trading hours. For this reason, we use E-mini S&P 500 futures (ES) rather than the cash index. Crude oil is represented by the NYMEX West Texas Intermediate contract (CL), which is physically delivered in Cushing, Oklahoma. It therefore reflects US oil prices, and thus the energy costs of S&P 500 firms, more closely than the North Sea Brent benchmark, which serves mainly as a reference for European and international crude prices. Natural gas (NG) is the NYMEX Henry Hub contract, the reference price for the US gas market. Gold (GC) captures demand for a safe haven in periods of financial stress (Baur & McDermott, 2010), and corn (C), used in the HAR, VAR and LVAR, adds an agricultural market not directly tied to energy and financial conditions. The ten-year Treasury yield serves as the benchmark for long-term US interest rates, which affect firms' discount rates and financing costs.

Daily RV5 and closing prices come from the Volatility Archive for Realized Estimates (VOLARE; Cipollini et al., 2026), which computes realised measures from intraday mid-quote prices supplied by Kibot. All realised variances enter in natural logarithms, which compresses their pronounced spikes. The forecasting target is equity log realised variance. The LHAR and LVAR additionally use equity returns computed from closing prices to construct downside controls. Variable definitions and descriptive statistics are reported in Table A2.

Our analysis uses the final shared dataset "vol_data_futures_yield_final_version.csv", prepared from the VOLARE and FRED data.

The Treasury yield (DGS10) is published by the Board of Governors of the Federal Reserve System and obtained from FRED. As no intraday bond data are available, squared and absolute yield changes, measured in basis points squared and basis points respectively, serve as proxies for Treasury-market variation, although they are not directly equivalent to the intraday futures measures. Because yields are reported in whole basis points, the change is exactly zero on many days.

The day-to-day basis point change is defined as:

$$\Delta y_t = \left(DGS10_t - DGS10_{t-1}\right) \times 100$$

These genuine zeros are retained, so both proxies remain unlogged. Incomplete dates are excluded rather than filled in. Treasury changes are calculated after aligning the series on the common calendar and therefore measure changes between consecutive retained observations. An additional one-observation Treasury delay is examined separately as a sensitivity check in the VAR analysis.

The VOLARE data start in September 2009, but the early gold series contains missing observations and implausible extreme values, hence we start in January 2011. The common sample contains 3,910 dates from 4 January 2011 to 31 August 2026. The first 3,127 observations, up to 12 July 2023, constitute the initial training history, and the remaining 783 form the evaluation sample for the VAR and LVAR comparisons (Table A1 and Figure A1). Model-specific lag and rolling-window construction reduces the usable estimation sample. We preserve this cutoff in the VAR and LVAR comparisons, so these specifications face the same evaluation targets. The original HAR and LHAR outputs use separate 80/20 splits, producing 778 and 774 evaluation targets respectively; common-date HAR/LHAR benchmarks are calculated separately for comparisons across model families. We do not remove further observations, so crisis episodes such as April 2020 remain in the sample.

Source: Andersen, T. G., Bollerslev, T., Diebold, F. X., & Labys, P. (2003). Modeling and Forecasting Realized Volatility. Econometrica, 71(2), 579–625. http://www.jstor.org/stable/3082068

3. Empirical Methodology

We examine whether information from other markets improves forecasts of equity realised variance using three complementary approaches: a heterogeneous autoregressive (HAR) model, a vector autoregressive (VAR) model and a leveragerd VAR model. These approaches allow us to investigate the predictive contribution of additional market information under different model specifications. The following subsections present each approach and explain the main modelling choices.

3.1 The Heterogeneous Autoregressive (HAR) Model

Leveraging high-frequency intraday data to measure realized volatility, we employ the Heterogeneous Autoregressive (HAR-RV) model of Corsi (2009). Unlike daily GARCH or short-memory ARMA specifications, which struggle to capture the slow, hyperbolic decay of volatility (long memory), the HAR framework exploits intraday precision to model volatility dynamics across distinct horizons.

The HAR framework circumvents these limitations by building upon the Heterogeneous Market Hypothesis. This hypothesis posits that financial markets are populated by distinct agent profiles operating across divergent temporal horizons: short-term intraday market participants and high-frequency traders who monitor and react to daily volatility shocks; medium-term institutional rebalancers and swing traders who condition their exposure on weekly market variance; and long-term fundamental asset managers and pension funds whose capital allocations depend on monthly macro volatility trends.

The standard HAR process captures long memory through a simple linear combination of past realized volatility across three intuitive horizons: daily ($d = 1$), weekly ($w = 5$), and monthly

($m = 22$ trading days). Formally, the baseline model predicting next-day realized volatility is specified as:

$$\ln RV_{t+1}^{(d)} = \beta_0 + \beta_d \ln RV_t^{(d)} + \beta_w \ln RV_t^{(w)} + \beta_m \ln RV_t^{(m)} + \epsilon_{t+1}$$

where:

$$\ln RV_t^{(w)} = \frac{1}{5}\sum_{i=0}^{4} \ln RV_{t-i}^{(d)} \quad \text{and} \quad \ln RV_t^{(m)} = \frac{1}{22}\sum_{i=0}^{21} \ln RV_{t-i}^{(d)}$$

represent the moving averages over weekly and monthly horizons.

Incorporating Asymmetries: The Leverage LHAR Model

Because negative equity returns induce a disproportionate surge in future volatility, a phenomenon attributed to financial leverage effects, we augment the baseline HAR to an LHAR-RV framework following Corsi and Renò (2012). We define the daily negative return shock as:

$$r_t^- = \min(r_t, 0)$$

where $r_t = \frac{\text{ret}_t}{100}$ denotes the daily percentage returns expressed in decimals. Multi-horizon negative return cascades are constructed identically:

$$r_{t,w}^- = \frac{1}{5}\sum_{i=0}^{4} r_{t-i}^-, \quad r_{t,m}^- = \frac{1}{22}\sum_{i=0}^{21} r_{t-i}^-$$

These asymmetric downside terms are appended directly to the regressor matrix to isolate market-wide panic spillovers.

3.2 Vector autoregression

To identify the contribution of other markets, we compare VAR forecasts with an equity-only AR containing the same number of lags and calendar controls. This comparison separates cross-market information from the predictive value of equity volatility's own history. A persistence forecast, equal to the previous retained observation, provides an additional benchmark.

The dependent variable is log RV5 of E-mini S&P 500 futures. We estimate four VAR specifications: equity with oil and gold; equity with natural gas and the Treasury proxy; equity with all four commodities, including corn; and a full system containing all six series. Each equation is estimated by OLS with an intercept, 11 month indicators and four weekday indicators. The calendar controls accommodate recurring seasonal differences without pre-filtering the data. Treasury variation enters as the supplied squared yield-change proxy. Appendix A describes the variables and common sample.

The initial training sample contains 3,127 observations through 12 July 2023. Lag selection uses only this sample: BIC selects four lags and AIC selects nine from a maximum of 20 in the full system. Four lags form the main specification; nine provide a sensitivity check. We

hold the lag order fixed across models within each comparison and re-estimate on an expanding sample, producing 783 one-step forecasts from 13 July 2023 to 31 August 2026. Forecasts use earlier retained observations and assume that all inputs dated before the target are available. Appendix B reports the specification checks.

Table 1 Out-of-sample accuracy with four lags

Model	MSE	MAE	MSE difference vs AR (%)
Persistence	0.43574	0.50428	+22.16
AR	0.35669	0.45396	0.00
VAR: oil and gold	0.35786	0.45575	+0.33
VAR: gas and treasury	0.35838	0.45651	+0.48
VAR: all commodities	0.35897	0.45759	+0.64
Full VAR	0.35918	0.45785	+0.70

Notes: 783 common forecast targets; errors are measured in equity log RV5. All AR/VAR models include identical calendar controls. Source: own calculations using the group dataset.

The equity AR reduces MSE by 18.14% relative to persistence. None of the four-lag VARs improves on the matched AR in either MSE or MAE. The full VAR's MSE is 0.70% higher, while adding oil and gold raises it by 0.33%. The improvement over persistence therefore comes from modelling equity's own dynamics and calendar effects; it cannot be attributed to the additional markets.

The nine-lag oil-and-gold VAR achieves a small 0.14% MSE reduction relative to AR(9), but its MAE is higher. Broader systems again fail to improve MSE. First differencing and an additional Treasury delay also leave the AR–VAR comparison unchanged (Appendix B). Mixed stationarity evidence and remaining residual autocorrelation warrant caution with conventional Granger-test inference. Overall, additional markets provide little incremental forecast value for this equity benchmark over the selected evaluation period. This finding does not establish that spillovers are absent or that the same result holds for other equity markets or forecast horizons.

3.3 Leverage-augmented VAR (LVAR)

Realised variance measures how much prices move but does not show whether the market ends higher or lower. Two days can have similar variance even if equity prices rise on one and fall on the other. We therefore ask whether recent equity price falls help predict future equity variance. We add three measures of negative ES returns to a standard VAR and call this the "leverage-augmented VAR" (LVAR). We then test whether commodity information adds further value.

$$z_{t+1} = c + \sum_{j=1}^{p} A_j z_{t+1-j} + \Gamma d_t + u_{t+1}, \qquad p = 5. \tag{1}$$

Here, z_t contains ES log RV5 and the log variances of the included commodities. The matrices A_j give the weights on the previous five variance observations, while Γ gives the weights on the three return measures in d_t. The constant is c, and $u_{t+1}$is the part the model does not explain. We estimate the weights by ordinary least squares (OLS), separately for each market equation, following the standard VAR framework (Lütkepohl,2005). The weights are learned from the data: a negative return does not automatically raise the variance forecast.

The leverage effect describes the tendency for negative equity returns to be associated with higher future variance. Following the idea of persistent leverage in Corsi and Renò (2012), we measure negative ES returns over the previous 1, 5 and 22 observations. For decimal close-to-close log returns r_t, these

measures are

$$d_t = (d_t^{(1)}, d_t^{(5)}, d_t^{(22)})', \qquad d_t^{(h)} = \left| \min\left(\frac{1}{h} \sum_{i=0}^{h-1} r_{t-i}, 0 \right) \right|. \tag{2}$$

Each measure is zero when the average return is positive or zero. When the average is negative, we use its magnitude. For example, an ES return of -2% gives a daily measure of 0.02, while a return of $+2$ gives zero. The supplied percentage returns are divided by 100. For the 5- and 22-observation measures, we first average the signed returns, allowing gains and losses to offset each other, and then measure the size of a negative average. These three inputs enter every equation and use only returns observed before the forecast target.

To measure what the return information adds, we compare LVAR with a plain VAR that uses the same variance lags, training observations and forecast dates. The plain VAR leaves out Γd_t. We also compare models with commodities against an equity-only model that keeps the same return measures. This separates the benefit of negative-return information from the benefit of adding commodities. The equity-only versions are an AR and an AR with downside controls. Unlike the VAR in Section 3.2, these models have no Treasury or calendar controls and use five rather than four variance lags. We therefore use our own plain benchmark to measure the downside contribution.

We use the final dataset described in Section 2. Building the lagged variables and return averages leaves 3,867 estimation outcomes, including 3,084 initial training outcomes through 12 July 2023. BIC on the training-only five-market plain VAR selects five lags from candidates 1-20. Each candidate uses the same outcomes. We keep five lags for all main VAR and LVAR comparisons. AIC selects ten, which we examine separately. The 1/5/22 return windows are fixed, as in LHAR, rather than selected by BIC.

We produce 783 forecasts from 13 July 2023 to 31 August 2026. Before each forecast, the model is fitted using only observations available at that point. Once the next outcome is observed, it is added to the training history for later forecasts. This expanding procedure never uses future commodity values or future equity returns.

Model	ES Coefficients	MSE	MAE
Equity-only VAR(5)	6	0.368531	0.464175
Equity-only VAR(5) + downside controls	9	0.334215	0.442837
LVAR: oil and gold	19	0.336611	0.443919
LVAR: natural gas	14	0.332670	0.442343
LVAR: oil, natural gas and gold	24	0.334549	0.443412
LVAR: all four commodities	29	0.334687	0.443561

Adding downside controls to the equity-only AR(5) lowers MSE from 0.368531 to 0.334215, an improvement of 9.31%. The return measures therefore provide useful forecast information beyond past equity variance in this comparison. Adding oil and gold to the model with downside controls raises MSE by 0.72%, and adding all four commodities raises it by 0.14%. Natural gas alone lowers MSE by 0.46%. These commodity differences are small. The descriptive uncertainty intervals in the appendix include zero for the commodity comparisons, while the downside improvement remains positive across the checks. We find a clearer benefit from equity-return information than from additional commodities.

Joint coefficient tests show that some commodity histories help explain the observations used to estimate the model. This does not mean they improve forecasts of later observations. The results do not establish that commodities and equities are unrelated; they show limited additional forecasting value in these models and over this evaluation period.

There are also reasons to interpret the results cautiously. ADF and KPSS give conflicting evidence about stationarity. The estimated VAR dynamics pass the stability check, but the residuals still show dependence over time, so the model does not capture every pattern in the data. We use standard errors that allow for autocorrelation and changes in error variance (HAC with bandwidth 22, Newey and West, 1987). This adjusts the reported uncertainty, but does not solve the stationarity issue or missing model dynamics. With ten lags, the equity-only model with downside controls has MSE 0.328510, and adding oil and gold still does not improve its forecast accuracy. Because the evaluation period has already been examined, these findings remain exploratory. We

do not test causal effects or whether the forecasts improve portfolio or risk-management decisions.

4. Empirical results

The empirical results distinguish improvements obtained from modelling equity's own history from those obtained by adding information from other markets. Across the completed HAR and VAR analyses, the additional markets produce small and specification-dependent changes in forecast accuracy. The largest improvement within the HAR framework comes from incorporating negative equity returns.

The baseline HAR achieves an out-of-sample MSE of 0.3639. None of its commodity extensions improves on this benchmark: their MSEs range from 0.3640 to 0.3649. Although some extensions improve in-sample fit, these improvements do not translate into better forecasts. For example, adding all four commodities increases R^2 from 0.7171 to 0.7191, while MSE rises to 0.3649. The results therefore provide no evidence that the commodity variables improve the baseline HAR's forecasting accuracy in the evaluation sample.

Adding negative-return terms produces a substantially larger improvement. The baseline LHAR reduces MSE from 0.3639 to 0.3254, approximately 10.6%, while adjusted R^2 increases from 0.7169 to 0.7442. This suggests that negative equity returns contain useful forecasting information beyond past realised variance. The forecast comparison alone, however, does not establish the causal mechanism behind this improvement.

Commodity information contributes modest gains once these return terms are included. The LHAR model with oil and natural gas achieves an MSE of 0.3246, approximately 0.25% below the equity-only LHAR. Adding the Treasury variation proxy lowers MSE further to 0.3241. This is the lowest reported MSE among the HAR-based specifications, but represents an improvement of only approximately 0.40% over the baseline LHAR. Most of the improvement relative to the original HAR therefore comes from the negative equity-return terms.

The information criteria and forecast results also favour different specifications. Among the reported LHAR models, the three-commodity specification with Treasury variation has the lowest AIC, whereas the oil-and-natural-gas specification with Treasury variation has the lowest forecast MSE. BIC favours the equity-only baseline within both the HAR and LHAR families. Consequently, improvements in historical fit should not be interpreted as evidence of superior out-of-sample performance.

Model	Out-of-sample MSE
Baseline HAR	0.3639
HAR-X: oil and natural gas	0.3643
Baseline LHAR	0.3254
LHAR-X: oil and natural gas	0.3246
LHAR-X: oil, natural gas and Treasury variation	0.3241

Persistence benchmark for VAR analysis	0.4357
Equity-only AR(4)	0.3567
VAR(4): oil and gold	0.3579
Full VAR(4)	0.3592

The VAR results reinforce the importance of a suitable equity-only benchmark. The four-lag AR reduces MSE by 18.14% relative to persistence. However, every four-lag VAR has higher MSE and MAE than this matched AR. Adding oil and gold increases MSE by 0.33%, while the full system increases it by 0.70%. The improvement over persistence therefore reflects the modelling of equity's own dynamics and calendar effects, rather than the inclusion of other markets.

The nine-lag sensitivity check gives a marginally different result: the oil-and-gold VAR reduces MSE by 0.14% relative to AR(9), but has higher MAE. The broader systems again fail to improve MSE. These results do not establish a consistent forecasting benefit from additional markets.

[Leveraged VAR results to be added: report its equity-only benchmark, the changes produced by adding other markets, and whether the conclusions hold across the specifications examined. Distinguish the contribution of the leverage terms from the contribution of cross-market variables.]

Taken together, the completed analyses suggest that modelling equity's own volatility and return history contributes more to forecast accuracy than adding the market variables examined here. Some LHAR extensions deliver small MSE reductions, while the baseline HAR and four-lag VAR extensions worsen accuracy. These are forecasting findings for the selected sample and specifications; they do not establish the absence of economic connections or volatility spillovers between markets.

5. Economic interpretation

The empirical results suggest that incorporating information from commodity and Treasury markets provides limited additional economic value when forecasting S&P 500 futures volatility. The equity-only AR model achieves a mean squared error (MSE) of 0.356682, compared with 0.435742 for the persistence benchmark, representing an improvement of approximately 18.14%. In contrast, adding oil and gold increases the MSE by 0.33%, while including all four additional markets increases it by 0.66%. These relatively small differences indicate that incorporating cross-market information does not meaningfully improve forecasting accuracy.

From an economic perspective, these findings suggest that historical equity volatility already captures much of the information relevant for predicting future market uncertainty. Although commodity and equity markets are interconnected through common macroeconomic shocks, inflation expectations, and changes in investor sentiment, these relationships do not necessarily translate into additional predictive value.

The findings also have practical implications for financial institutions and investors, who rely on volatility forecasts for risk management, portfolio allocation, and Value-at-Risk calculations. Given the limited improvements in forecasting accuracy, the additional complexity and data requirements of multivariate models may outweigh their practical benefits. Consequently, simpler models relying on historical equity volatility may offer a more efficient forecasting approach.

Overall, the results highlight the economic importance of volatility persistence while suggesting that incorporating additional market information provides limited incremental forecasting value within the specifications and evaluation period examined.

Consistent with Corsi (2009), who finds that a simple autoregressive model of realized volatility forecasts S&P 500 futures volatility well, the equity market's own history appears to capture most of the relevant information. Two mechanisms may explain why the linkages between markets do not translate into forecasting gains. First, common shocks are likely to show up in equity variance on the same day, so lagged commodity or bond information adds little to lagged equity variance. Second, transmission may run mainly from equities to other markets. Diebold and Yilmaz (2012) find that volatility spillovers across US stock, bond, foreign exchange and commodity markets were limited before the 2007 crisis. As the crisis intensified, the spillovers ran particularly from the stock market to the other markets. Predictive gains from outside the stock market also tend to be episodic. Paye (2012) finds that forecasts based on macroeconomic variables struggle to outperform a univariate benchmark out-of-sample, with predictive power concentrated around the onset of recessions. Our evaluation period (July 2023 to August 2026) contains no recession or crisis comparable to 2008 or 2020, so such episodes have little opportunity to appear.

Banks and other financial institutions use volatility forecasts, for example, to calculate Value-at-Risk, which rises and falls with the forecast volatility. A forecast that changes MSE by less than 1% would therefore hardly change the resulting Value-at-Risk, whereas modelling the equity market's own history improves accuracy by 18% relative to the persistence benchmark. Such accuracy matters most at short horizons like the one-day horizon studied here. Christoffersen and Diebold (2000) show that volatility forecastability decays quickly with the horizon and is mainly relevant for short-horizon risk management. We evaluate statistical accuracy only, however, and do not test whether the forecasts lead to better risk-management decisions.

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6. Robustness and limitations

Several limitations qualify the interpretation of the forecasting results across the HAR, VAR and LVAR approaches.

First, Treasury-market variation is measured using squared changes in the ten-year US yield, based on daily observations, rather than intraday realised variance. Although bond prices and yields are closely linked, this proxy captures net yield movements and misses fluctuations that reverse within the observation interval. It therefore differs from the RV5 measures used for equity and commodity futures. The Treasury results concern this particular proxy and do not

establish whether Treasury futures realised variance would provide additional forecasting information.

Second, the analysis concerns S&P 500 futures and should not be generalised to equity markets as a whole. Market liquidity, sector composition and exposure to commodity prices may affect the usefulness of cross-market information. Oil volatility could provide greater predictive value for indices with larger energy-sector weights or for economies more dependent on oil production or consumption. Less liquid markets may also incorporate information at a different speed.

Third, the daily forecasting framework may obscure relationships operating within the trading day. Although RV5 is constructed from intraday returns, the models use daily variance observations. Common news may affect equity, commodity and Treasury markets almost simultaneously, allowing equity's own observed volatility to reflect much of the information available in other markets before the forecast is made. Limited gains from lagged cross-market variables would therefore be compatible with economically important connections between markets. Our analysis does not identify the timing or direction of such transmission.

Fourth, forecast performance depends on the sample period, horizon and specifications considered. Average accuracy may conceal differences across periods of financial stress and normal market conditions. Comparing HAR, VAR and LVAR models broadens the analysis, but does not cover every possible specification or allow unrestricted changes in the underlying relationships. Information that contributes little in one setting could remain useful at another horizon or under different economic conditions.

Finally, the statistical diagnostics call for caution. Stationarity tests provide mixed evidence, and the residual autocorrelation documented for the VAR specifications limits confidence in conventional coefficient and Granger tests. These findings should not be assumed to apply identically to every model, but they reinforce the need to qualify the conclusions. Small differences in historical forecast accuracy do not establish general model superiority. The analysis also does not assess whether the forecasts improve portfolio allocation or risk-management decisions.

7. Conclusion

8. Division of labour

Task	*Lead*	*Supporting contributor*
Research question and economic motivation	**Amina**	**Femke**

Data	**Femke**	**Amina**
HAR model	**Dean**	**Amina**
VAR model	**Tim**	**Femke**
LVAR model	**Jake**	**Dean**
Empirical results	**All team members**	**—**
Economic interpretation	**Femke**	**Amina**
Robustness and limitations	**Tim**	**—**

Empirical appendix

Appendix A Data and preliminary checks

The supplied group dataset contains 3,910 dated observations and 24 columns. Dates are unique and ordered, and all numeric entries are finite and non-missing. The VAR analysis retains every row and selects five existing futures log-RV5 columns plus the Treasury variation proxy. Stored log columns equal the natural logarithm of their corresponding RV5 columns; no additional log transformation is applied.

Table A1 Common observation sample

Sample	Dates	Observations
Full panel	4 January 2011–31 August 2026	3,910
Initial training	4 January 2011–12 July 2023	3,127
Test targets	13 July 2023–31 August 2026	783

The existing training cutoff is retained. Counts precede model-specific lag losses. The expanding procedure incorporates each observed test outcome only into subsequent forecasts.

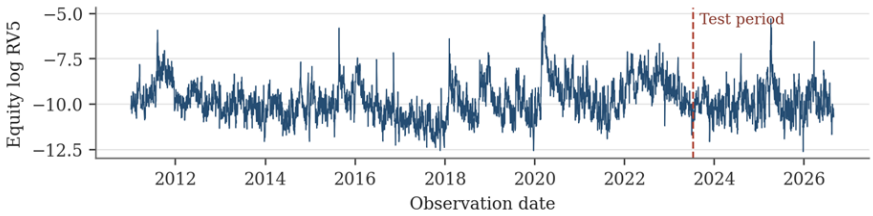
Table A2 Variables and initial training descriptives

Series	CSV column	Mean	Std. deviation
Equity	log_rv5_ES	-9.761	1.053
WTI oil	log_rv5_CL	-8.015	0.945
Gold	log_rv5_GC	-9.604	0.730
Natural gas	log_rv5_NG	-7.409	0.860
Corn	log_rv5_C	-8.693	0.821
Treasury proxy	rv_yield_10	26.749	53.987

Futures variables are natural log RV5. Treasury variation is in basis points squared, unlogged; 308 full-sample observations are zero. Units differ across these two representations.

The Treasury proxy equals the squared basis-point change in the yield between successive retained dates, wherever the change can be verified within the file. It can therefore span a gap in the common calendar and is not intraday bond realised variance. The baseline uses its ordinary model lags and assumes previous-date information is available; an extra-delay check appears in Appendix B. No values are interpolated or seasonally pre-filtered.

Figure A1 Equity log realised variance and the evaluation period



The dashed line marks the fixed training cutoff. The plot displays the full supplied panel for description; preliminary tests use only the initial training sample.

Stationarity and seasonality

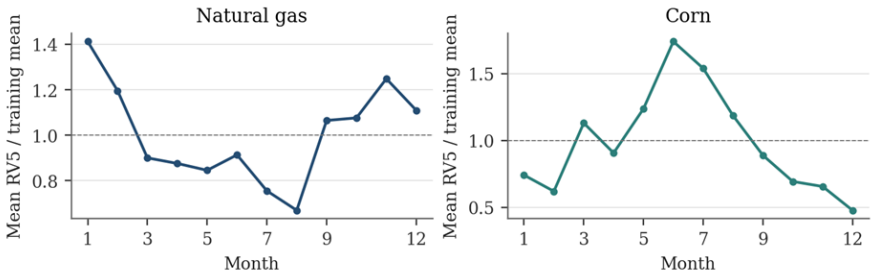
Table A3 Stationarity tests and persistence in the initial training sample

Series	ADF statistic	KPSS statistic	Lag-one ACF
Equity	-5.661	0.896	0.855
WTI oil	-4.152	2.133	0.874
Gold	-5.136	1.041	0.667
Natural gas	-3.757	2.857	0.792
Corn	-4.825	0.883	0.619
Treasury proxy	-8.132	1.034	0.213

ADF: intercept, no trend, 20 lagged differences; 5% critical value approximately −2.86. KPSS: level-stationarity null and 28 Bartlett lags; 5% critical value 0.463. Tests use the series as entered, before calendar controls. ACF denotes sample autocorrelation.

ADF rejects a unit root for every series, while KPSS rejects constant-mean level stationarity for every series. This mixed evidence prevents an unqualified claim that the data are stationary. Strong persistence, changing volatility regimes and deterministic patterns may contribute to the disagreement. Log levels are retained as the baseline forecasting specification, while the first-difference check in Appendix B assesses whether the AR–VAR comparison depends on this choice.

Figure A2 Monthly volatility patterns in natural gas and corn



Each point is the arithmetic mean of unlogged RV5 for that month, divided by the series' overall initial-training mean. These are descriptive monthly averages, not causal estimates or formal tests of seasonality.

Natural gas has its highest monthly mean in January and its lowest in August. Corn peaks in June; July's mean RV5 is 3.21 times December's. Rejecting a unit root does not remove these calendar patterns. The AR and VAR equations therefore include 11 month and four weekday indicators, estimated within each expanding training sample. The underlying observations remain unchanged, and seasonal coefficients are never estimated using future test outcomes.

Appendix B VAR specification and checks

All systems include equity log RV5. Additional futures variables are their log RV5 series; Treasury variation is the supplied squared yield change in basis points squared. Every equation contains an intercept and 15 calendar indicators. The equity-only AR uses the same deterministic controls. Models are estimated by OLS, equation by equation.

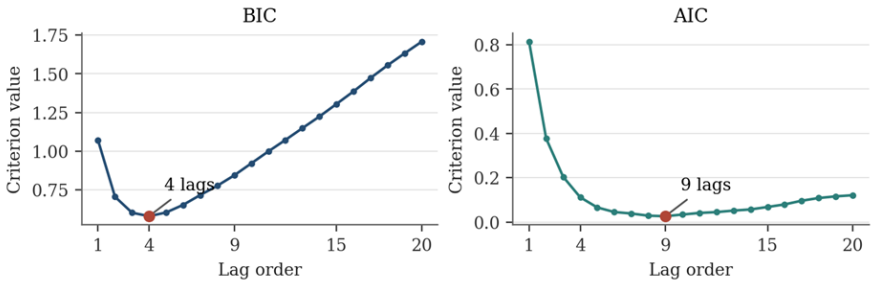
Table B1 Initial training specifications with four lags

Model	Variables	Coefficients per equation	Largest root	Serial-correlation p-value
AR	ES	20	0.938	<0.001
Oil and gold	ES, CL, GC	28	0.969	<0.001
Gas and Treasury	ES, NG, TY	28	0.966	<0.001
All commodities	ES, CL, GC, NG, C	36	0.975	<0.001
Full VAR	ES, CL, GC, NG, C, TY	40	0.975	<0.001

ES: equity; CL: oil; GC: gold; NG: natural gas; C: corn; TY: Treasury proxy. Coefficient counts include the intercept and calendar indicators. Roots are companion-matrix eigenvalue moduli. Serial-correlation checks are Breusch–Godfrey LM tests with 20 residual lags, applied to the equity equation.

All initial training estimates satisfy the stability condition, with roots inside the unit circle. However, the residual tests reject absence of serial correlation. Increasing to nine lags reduces but does not eliminate this problem: the full VAR's residual-test p-value is 0.010. The models are therefore evaluated primarily on out-of-sample forecasting, with conventional in-sample test results interpreted cautiously.

Figure B1 Lag selection in the initial training sample



AIC and BIC are computed for the full six-variable system using the same 3,107 regression observations at every candidate order. BIC selects four lags and AIC nine. After selection, estimation uses the available sample for that order: 3,123 initial regression rows at four lags and 3,118 at nine. No test-period outcomes enter lag selection.

Forecast sensitivity and in-sample predictability

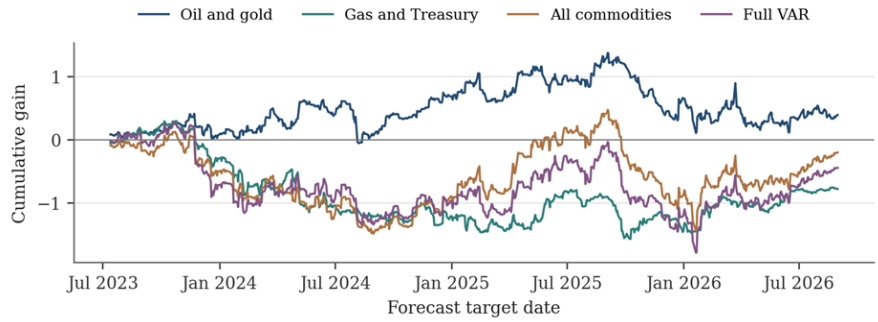
Table B2 Forecast accuracy and joint Granger tests

Model	MSE 4 lags	MSE 9 lags	Granger p 4 lags	Granger p 9 lags
AR	0.35669	0.35713	—	—
Oil and gold	0.35786	0.35662	0.040	0.382
Gas and Treasury	0.35838	0.35812	0.057	0.399
All commodities	0.35897	0.35738	0.028	0.539
Full VAR	0.35918	0.35769	0.086	0.726

MSE uses the same 783 test targets. Granger tests use only the initial training sample and jointly set all additional-market lag coefficients in the equity equation to zero, conditional on equity lags and calendar controls. Reported p-values are conventional F-test values; residual serial correlation limits their reliability.

At four lags, the oil-and-gold and all-commodity systems yield nominal p-values below 0.05. Neither finding persists at nine lags, and neither system improves four-lag forecast accuracy. These results illustrate why apparent in-sample predictability is insufficient to establish a forecasting gain. They do not identify structural causal effects.

Figure B2 Cumulative forecast gain over AR with nine lags



Cumulative gain equals the sum of AR squared errors minus VAR squared errors. Positive values favour the VAR. All models use nine lags and the same controls. The small final oil-and-gold advantage varies over the evaluation period.

Two further checks retain all 783 targets. First differencing, with four lags selected by training-sample BIC, gives level-forecast MSEs of 0.37291 for the AR and 0.37312 for the full VAR. Adding one further retained-observation delay to the Treasury proxy gives 0.35669 for a sample-matched AR and 0.35915 for the full VAR. Both checks leave the main AR–VAR ranking unchanged.

Appendix C: HAR Model

Inclusion of Exogenous Variables

To test for cross-asset volatility spillovers, the baseline model is augmented with exogenous commodity markets and macroeconomic yield volatility. The exogenous regressors are incorporated at day t to forecast next-day equity index volatility ($\ln RV_{t+1}^{ES}$):

$$\ln RV_{t+1}^{ES} = \beta_0 + \mathrm{HAR}_t^{ES} + \mathrm{Leverage}_t^{ES} + \sum_{k \in \mathcal{X}} \beta_k^{\top} \mathrm{X}_{k,t} + \beta_{\mathrm{yield}} Y_t + \epsilon_{t+1}$$

1. Multi-Horizon Commodity Regressors

For each exogenous commodity $k \in \{\text{Crude Oil (CL)}, \text{Natural Gas (NG)}, \text{Gold (GC)}, \text{Corn (C)}\}$, we compute realized volatility from 5-minute intraday prices and take its natural logarithm:

$$\ln RV_{k,t}^{(d)} = \ln\left(RV_{k,t}\right)$$

Following the heterogeneous market framework, each commodity enters the model decomposed into daily, weekly, and monthly moving averages:

$$\mathrm{X}_{k,t} = \begin{bmatrix} \ln RV_{k,t}^{(d)} \\ \ln RV_{k,t}^{(w)} \\ \ln RV_{k,t}^{(m)} \end{bmatrix}, \quad \text{where} \quad \ln RV_{k,t}^{(w)} = \frac{1}{5}\sum_{i=0}^{4} \ln RV_{k,t-i}^{(d)}, \quad \ln RV_{k,t}^{(m)} = \frac{1}{22}\sum_{i=0}^{21} \ln RV_{k,t-i}^{(d)}$$

This multi-horizon formulation allows the model to capture both immediate intraday commodity shocks and persistent, longer-term variance trends.

To evaluate how yield fluctuations transmit into equity volatility, two competing proxies are tested as 1-day exogenous regressors (Y_t):

- Realized Variance Proxy (Squared Deviations): $Y_t = (\Delta y_t)^2$
- Absolute Deviation Proxy: $Y_t = |\Delta y_t|$

All exogenous regressors are aligned at time t with standard HAC robust covariance estimation (Newey-West with 22 lags) to account for residual serial correlation across rolling horizons.

Empirical Findings

Base models:

Model Specification	R^2	Adj. R^2	AIC	BIC	MSE
Baseline HAR	0.7171	0.7169	6296.422	6321.485	0.3639
HAR-X (Oil)	0.7175	0.7171	6296.433	6340.293	0.364
HAR-X (NatGas)	0.7175	0.7171	6296.406	6340.266	0.3644
HAR-X (Gold)	0.7179	0.7174	6291.889	6335.748	0.3649
HAR-X (Corn)	0.7172	0.7168	6300.78	6344.639	0.364
HAR-X (Energy: Oil + NG)	0.718	0.7174	6295.856	6358.513	0.3643
HAR-X (Energy: Corn + Gold)	0.718	0.7174	6295.758	6358.415	0.3648
HAR-X (3 Comms: CL, NG, GC)	0.719	0.7181	6288.39	6369.844	0.3648
HAR-X (4 Comms: CL, NG, GC, C)	0.7191	0.718	6293.147	6393.397	0.3649
Baseline LHAR	0.7446	0.7442	5887.161	5930.982	0.3254
LHAR-X (Oil)	0.745	0.7444	5887.296	5949.898	0.3251
LHAR-X (NatGas)	0.7455	0.7449	5879.92	5942.522	0.3249
LHAR-X (Gold)	0.7455	0.7449	5880.904	5943.506	0.3262
LHAR-X (Corn)	0.7448	0.7442	5891.324	5953.927	0.3251
LHAR-X (Energy: Oil + NG)	0.7459	0.7451	5880.038	5961.421	0.3246
LHAR-X (Energy: Corn + Gold)	0.7456	0.7448	5884.174	5965.557	0.3258
LHAR-X (3 Comms: CL, NG, GC)	0.7467	0.7457	5873.764	5973.928	0.3252
LHAR-X (4 Comms: CL, NG, GC, C)	0.7468	0.7457	5877.756	5996.7	0.3249

Models including yield volatility

LHAR (Energy: CL, NG) + Yield Squared	0.7465	0.7456	5873.2	5960.843	0.3241
LHAR (3 Comms: CL, NG, GC) + Yield Squared	0.7471	0.7461	5869.469	5975.893	0.3248
LHAR (4 Comms: CL, NG, GC, C) + Yield Squared	0.7473	0.746	5873.221	5998.426	0.3244

Appendix D: LVAR specification and checks

This appendix supports the LVAR in Section 3.3. The main specification uses five variance lags, an intercept and the downside measures in Equation (2), without Treasury or calendar controls. Forecasts use the same 783 evaluation targets as Section 3.2, from 13 July 2023 to 31 August 2026. Appendix B covers Section 3.2's different VAR specification.

D.1 Lag selection and forecast sensitivity

AIC and BIC use the plain five-market system (equity, oil, corn, gold and natural gas), estimated only on the initial training sample through 12 July 2023. Each candidate uses the same 3,084 outcomes. BIC selects five variance lags; AIC selects ten. The downside windows stay fixed at 1, 5 and 22 observations.

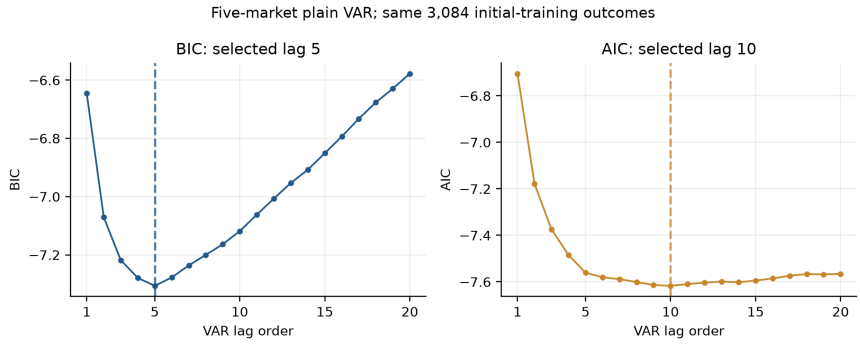


Figure D1: Lag selection for the initial-training five-market VAR

Table D1: forecast accuracy with five and ten variance lags

	MSE		MAE	
Model	5 lags	10 lags	5 lags	10 lags
Equity-only AR	0.368531	0.365940	0.464175	0.463306
Equity-only AR + downside controls	0.334215	0.328510	0.442837	0.440629
LVAR: oil and gold	0.336611	0.330827	0.443919	0.442912
LVAR: natural gas	0.332670	0.328684	0.442343	0.441537
LVAR: oil, natural gas and gold	0.334549	0.330157	0.443412	0.442896
LVAR: all four commodities	0.334687	0.330471	0.443561	0.442872

Notes: 783 common forecast targets and an expanding estimation sample. Lower values indicate smaller errors. The equity-only models are ARs; every LVAR includes the same downside controls. “All four” means oil, corn, gold and natural gas. Source: own calculations.

D.2 Stationarity and fitted-model checks

Table D2: Stationarity tests for the initial-training log-RV5 series

Series	ADF statistic	ADF p	KPSS statistic	KPSS p
Equity	-5.5373	< 0.001	0.8349	≤ 0.010
WTI oil	-3.9928	0.0014	1.9409	≤ 0.010
Gold	-5.1617	< 0.001	0.9606	≤ 0.010
Natural gas	-3.9078	0.0020	2.6295	≤ 0.010
Corn	-5.5897	< 0.001	0.8136	≤ 0.010

Notes: 3,106 observations per series, from 3 February 2011 to 12 July 2023. Both tests include an intercept and no trend. ADF selects its auxiliary lags by AIC; KPSS uses an automatic bandwidth. KPSS probabilities are bounded by its lookup table; values of 0.010 indicate $p \leq 0.010$. Probabilities below 0.001 are displayed as < 0.001. Source: own calculations.

ADF rejects a unit root for every series, while KPSS rejects level stationarity. This conflicting evidence prevents an unqualified claim that the series are stationary. The fitted-model checks below assess stability and unexplained patterns separately.

Table D3: Initial-training stability and equity-residual checks for LVAR(5)

Model	Largest root	Ljung–Box p	ARCH p
Equity-only AR + downside controls	0.9159	< 0.001	0.4184
LVAR: oil and gold	0.9669	< 0.001	0.3726
LVAR: oil, natural gas and gold	0.9777	0.0018	0.3227
LVAR: all four commodities	0.9786	0.0019	0.3126

Notes: 3,084 training outcomes. The largest root is the modulus of a companion-matrix eigenvalue, conditional on the downside controls. Values below one satisfy the variance-dynamics stability condition. Ljung–Box tests use 30 residual lags without a fitted-parameter degrees-of-freedom adjustment; ARCH tests use five lags. These are diagnostic checks. Source: own calculations.

The reported roots are below one, but the Ljung–Box results indicate remaining residual dependence. The ARCH tests do not reject their null at 5%. Passing these stability or ARCH checks does not establish stationarity or complete model adequacy.

Table D4: Joint tests of commodity-lag coefficients in the equity equation

Model	Coefficients tested	Wald statistic	HAC p
LVAR: oil and gold	10	37.7788	< 0.001
LVAR: oil, natural gas and gold	15	63.7299	< 0.001
LVAR: all four commodities	20	69.6887	< 0.001

Notes: the null sets all included commodity-lag coefficients in the equity equation to zero, conditional on equity lags and downside controls. Tests use 3,084 initial-training outcomes and HAC covariance with bandwidth 22. Reported probabilities use an asymptotic chi-squared reference distribution. Source: own calculations.

The joint tests indicate in-sample predictive information from the commodity histories. Mixed stationarity evidence and remaining residual dependence warrant caution with inference. These results do not establish causal effects or improvements in out-of-sample forecasting.

D.3 Uncertainty around forecast improvements

Figure D2 and Table D5 compare each candidate with its named benchmark on the same forecast dates. Positive gains indicate lower MSE for the candidate. To describe uncertainty, we resample the paired forecast losses in circular blocks of 20 observations, using 2,000 draws. Resampling blocks keeps some dependence between nearby observations. The intervals are descriptive and are not formal nested-model or selection-adjusted significance tests.

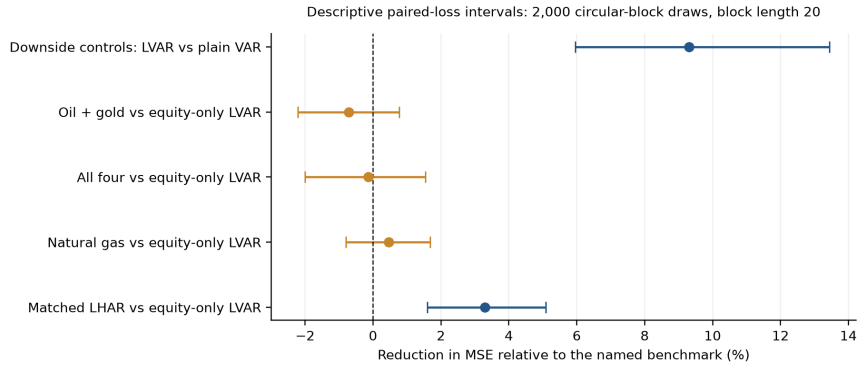


Figure D2: Incremental forecasting value of downside controls and commodities
Notes: points show percentage reductions in MSE relative to the named benchmark. Bars are descriptive 95% paired-loss intervals; the dashed line marks zero gain. The LHAR comparison is recomputed on the same 783 targets and differs from the original HAR notebook's separate 80/20 split. Source: own calculations.

Comparison	Gain (%)	Lower (%)	Upper (%)
Downside controls vs plain equity AR	9.31	5.95	13.44
Oil and gold vs equity-only downside model	-0.72	-2.21	0.77
All four commodities vs equity-only downside model	-0.14	-2.00	1.55
Natural gas vs equity-only downside model	0.46	-0.79	1.69
Matched equity-only LHAR vs equity-only downside model	3.30	1.59	5.09

Notes: all models use five variance lags, except LHAR, which uses fixed 1/5/22 variance averages. The gain is $100(1 - \mathrm{MSE}_{\mathrm{candidate}}/\mathrm{MSE}_{\mathrm{benchmark}})$. Bounds are the 2.5th and 97.5th percentiles from 2,000 circular-block draws with block length 20. Negative gains indicate higher MSE. Source: own calculations.

The downside-control interval is entirely positive, while each commodity interval includes zero. The downside interval also remains positive when block lengths of 5 and 60 are used. The common-date LHAR comparison shows a descriptive advantage over the equity-only model with downside controls. Together, these checks provide clearer evidence of forecasting value from equity-return information than from additional commodities in the reported specifications. Because the evaluation period has already been examined, the findings remain exploratory and do

not establish general model superiority or benefits for portfolio and risk-management decisions.

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Peter F. Christoffersen, Francis X. Diebold; How Relevant is Volatility Forecasting for Financial Risk Management?. *The Review of Economics and Statistics* 2000; 82 (1): 12–22. doi: https://doi.org/10.1162/003465300558597

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